

Problem 8

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Problem 1 We want to approximate the area under the curve using a right Riemann Sum with the given value of n . Write the sum in summation notation and evaluate it.

(a) $\sin(x)$, $\left[0, \frac{\pi}{2}\right]$, $n = 3$

Explanation. $\Delta x = \frac{b-a}{n} = \frac{\frac{\pi}{2} - 0}{3} = \frac{\pi}{6}$.

$$x_k^* = x_k = a + k\Delta x = 0 + \frac{\pi}{6}k = \frac{\pi}{6}k.$$

$$\begin{aligned} \sum_{k=1}^3 f(x_k^*)\Delta x &= \sum_{k=1}^3 \left(f\left(\frac{\pi}{6}k\right) \cdot \frac{\pi}{6} \right) \\ &= \frac{\pi}{6} \sum_{k=1}^3 \sin\left(\frac{\pi}{6}k\right) \\ &= \frac{\pi}{6} \left(\sin\left(\frac{\pi}{6}\right) + \sin\left(\frac{2\pi}{6}\right) + \sin\left(\frac{3\pi}{6}\right) \right) \\ &= \frac{\pi}{6} \left(\sin\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{3}\right) + \sin\left(\frac{\pi}{2}\right) \right) \\ &= \frac{\pi}{6} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} + 1 \right) \\ &= \frac{\pi}{6} \left(\frac{3}{2} + \frac{\sqrt{3}}{2} \right) \\ &= \frac{\pi}{12}(3 + \sqrt{3}) \end{aligned}$$

(b) $f(x) = x^2 - 9x + 18$, $[7, 10]$, $n = 6$

Explanation. $\Delta x = \frac{b-a}{n} = \frac{10-7}{6} = \frac{1}{2}$.

Problem 8

$$x_k^* = x_k = a + k\Delta x = 7 + \frac{1}{2}k.$$

$$\begin{aligned} \sum_{k=1}^6 f(x_k^*)\Delta x &= f(7 + \frac{1}{2}) \cdot \frac{1}{2} + f(7 + 1) \cdot \frac{1}{2} + f(7 + \frac{3}{2}) \cdot \frac{1}{2} + f(7 + 2) \cdot \frac{1}{2} + f(7 + \frac{5}{2}) \cdot \frac{1}{2} + f(7 + 3) \cdot \frac{1}{2} \\ &= f(7.5) \cdot \frac{1}{2} + f(8) \cdot \frac{1}{2} + f(8.5) \cdot \frac{1}{2} + f(9) \cdot \frac{1}{2} + f(9.5) \cdot \frac{1}{2} + f(10) \cdot \frac{1}{2} \\ &= \frac{1}{2} (f(7.5) + f(8) + f(8.5) + f(9) + f(9.5) + f(10)) \\ &= \frac{1}{2} ((7.5^2 - 9(7.5) + 18) + (8^2 - 9(8) + 18) + (8.5^2 - 9(8.5) + 18) + (9^2 - 9(9) + 18) + (9.5^2 - 9(9.5) + 18) + (10^2 - 9(10) + 18)) \\ &= \frac{1}{2} ((7.5^2 + 8^2 + 8.5^2 + 9^2 + 9.5^2 + 10^2) - 9(7.5 + 8 + 8.5 + 9 + 9.5 + 10) + 6(18)) \\ &= \frac{397}{8} \end{aligned}$$

Or in sigma notation:

$$\begin{aligned} \sum_{k=1}^6 f(a + k\Delta x)\Delta x &= \sum_{k=1}^6 \left(f\left(7 + \frac{1}{2}k\right) \cdot \frac{1}{2} \right) \\ &= \frac{1}{2} \sum_{k=1}^6 \left(\left(7 + \frac{1}{2}k\right)^2 - 9\left(7 + \frac{1}{2}k\right) + 18 \right) \\ &= \frac{1}{2} \sum_{k=1}^6 \left(49 + 7k + \frac{1}{4}k^2 - 63 - \frac{9}{2}k + 18 \right) \\ &= \frac{1}{2} \sum_{k=1}^6 \left(4 + \frac{5}{2}k + \frac{1}{4}k^2 \right) \\ &= \frac{1}{2} \left(4 \sum_{k=1}^6 1 + \frac{5}{2} \sum_{k=1}^6 k + \frac{1}{4} \sum_{k=1}^6 k^2 \right) \\ &= \frac{1}{2} \left(4(6) + \frac{5}{2} \cdot \frac{(6)(7)}{2} + \frac{1}{4} \cdot \frac{(6)(7)(13)}{6} \right) \\ &= \frac{1}{2} \left(24 + \frac{105}{2} + \frac{91}{4} \right) \\ &= \frac{1}{2} \cdot \frac{96 + 210 + 91}{4} = \frac{397}{8} \end{aligned}$$